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# Neuron-Level Circuit Distillation for Arithmetic

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## Abstract

1 VG: ES:

## 2 1 Introduction

3 Large language models (LLMs) with tens of billions of parameters exhibit impressive emergent  
4 arithmetic and symbolic reasoning abilities [Dubey et al., 2024]. However, deploying such massive  
5 models is often infeasible due to computational and memory constraints. Knowledge distillation  
6 seeks to compress these capabilities into smaller student models, but standard approaches primarily  
7 focus on matching output distributions (e.g., KL divergence). Consequently, distilled models often  
8 reproduce the correct answers by memorizing brittle, surface-level heuristics rather than inheriting  
9 the teacher’s robust underlying algorithmic mechanisms [Wadhwa et al., 2024].

10 Mechanistic interpretability offers a different lens for understanding model behavior: many reasoning  
11 tasks in transformers are implemented by sparse, functionally specialized subcircuits composed of  
12 particular neurons and attention heads [Elhage et al., 2021]. This motivates a compelling question:  
13 *can we identify these exact reasoning circuits and distill them directly into smaller models?* Recent  
14 work has demonstrated the viability of macro-level circuit distillation, aligning entire functionally  
15 critical layers to bridge the reasoning gap [Singhal et al., 2025]. In this work, we push this paradigm  
16 to a finer granularity to study *neuron-level circuit distillation* for arithmetic reasoning.

17 Instead of relying on broad functional alignment or complex latent clustering, we isolate the precise  
18 sub-network driving mathematical reasoning. We identify a sparse set of the top- $k$  most functionally  
19 critical neurons in the teacher model using a cosine similarity-based activation analysis across the  
20 problem set. We then train the student to align its internal computations with this specific neuronal  
21 subset using a composite objective of traditional distillation and Centered Kernel Alignment (CKA)  
22 [Kornblith et al., 2019].

23 To rigorously validate our identified circuits before distillation, we also introduce a control-adjusted  
24 ablation methodology. Naively zeroing out activations (standard ablation) conflates the loss of  
25 specific reasoning mechanisms with a general degradation of model capacity. We address this by  
26 measuring the functional importance of our discovered circuit against a baseline defined by the  
27 expected performance drop of ablating  $n$  random neurons. In this context, we make the following  
28 contributions:

29 **Neuron-Level Circuit Distillation.** We introduce a mechanism-level distillation pipeline that aligns  
30 the top- $k$  functionally critical neurons between teacher and student models. We demonstrate that  
31 targeting this specific sparse subcircuit achieves superior mechanism-level transfer compared to both  
32 standard KL distillation and random-neuron alignment controls.

33 **Robust Ablation Evaluation.** We propose a rigorous ablation framework that calculates the expected  
34 drop of zeroing out  $n$  random neurons. By benchmarking our discovered circuits against this  
35 randomized baseline, we isolate the distinct functional importance of the arithmetic circuit, proving it  
36 is uniquely responsible for reasoning rather than general network capacity.

37 **Circuit Discovery via Activation Consistency.** We show that simple, scalable metrics—specifically,  
 38 tracking the geometric consistency of neuron activations via cosine similarity across arithmetic  
 39 datasets—are sufficient to localize highly specialized reasoning subnetworks across both 1B and 8B  
 40 parameter models.

## 41 2 Related Work

42 **Knowledge Distillation.** Traditional knowledge distillation transfers behavior by matching teacher  
 43 and student output logits, sometimes augmented with intermediate representation matching [Hinton  
 44 et al., 2015]. While effective for general performance transfer, these approaches do not explicitly  
 45 preserve internal computational mechanisms. Techniques like Chain-of-Thought (CoT) distillation  
 46 incorporate teacher rationales to guide reasoning, but studies show students may still imitate textual  
 47 patterns without acquiring the underlying algorithmic computation [Wadhwa et al., 2024].

48 **Mechanistic Interpretability and Circuits.** Mechanistic interpretability aims to decompose neural  
 49 networks into interpretable components that implement specific functions. The transformer circuits  
 50 framework formalizes this perspective, identifying sparse subnetworks responsible for behaviors  
 51 including induction, modular addition, and arithmetic reasoning [Elhage et al., 2021, Nanda et al.,  
 52 2023]. Scalable methods approximate component importance and enable circuit localization in large  
 53 models [Nanda, 2023]. Most relevant to our work, Singhal et al. [2025] introduced circuit distillation  
 54 by aligning critical layers between models. We extend this by localizing and distilling at the highly  
 55 granular, neuron-specific level.

56 **Representational Similarity Metrics.** Mechanism-level transfer requires comparing internal repre-  
 57 sentations across models that differ in scale. Centered Kernel Alignment (CKA) has emerged as a  
 58 robust metric due to its invariance to orthogonal transformations and isotropic scaling, and its stability  
 59 in high-dimensional settings [Kornblith et al., 2019]. It provides a mathematically grounded means  
 60 of aligning the internal activation geometries of our top- $k$  discovered neurons during the distillation  
 61 process.

## 62 3 Methods

63 We propose a three-stage framework for circuit distillation in mathematical reasoning tasks. Given a  
 64 teacher and student model trained on arithmetic problems, we first identify a sparse set of functionally  
 65 critical neurons in the teacher via cosine similarity-based activation analysis, then compress these  
 66 neurons into compact cluster representations via K-means, and finally train the student to align with  
 67 these cluster representations through a composite distillation objective. Each stage is designed to be  
 68 independently interpretable and jointly optimized toward producing a student that emulates not only  
 69 the teacher’s output distribution, but its underlying computational mechanisms.

### 70 3.1 Top- $k$ Neuron Circuit Discovery

71 The first stage of our pipeline identifies sparse sets of neurons in the student and teacher models that  
 72 are functionally critical for the arithmetic problem class,  $\mathcal{D}$ , we are training on.  
 73 To collect neuron activations, we perform a forward pass over  $\mathcal{D}$  of the model and collect the post-  
 74 activation outputs of every MLP intermediate neuron across all the layers. Note, only one generation  
 75 is needed as all answers can be encoded in a single token. Formally, let  $a_{i,j}^{(l)} \in \mathbb{R}^T$  denote the  
 76 activation vector of neuron  $j$  in layer  $l$  for problem  $i$  across a constant  $T$  tokens. Then, we score each  
 77 neuron by its mean pairwise cosine similarity across all  $\binom{|\mathcal{D}|}{2}$  problem pairs with clamping:

$$s_j^{(l)} = \frac{1}{\binom{|\mathcal{D}|}{2}} \sum_{1 \leq i < i' \leq |\mathcal{D}|} \frac{a_{i,j}^{(l)} a_{i',j}^{(l)}}{\|a_{i,j}^{(l)}\|_\epsilon \|a_{i',j}^{(l)}\|_\epsilon} \quad \|x\|_\epsilon := \max(|x|, \epsilon) \quad (1)$$

78 Therefore, we can interpret that a neuron with high  $s_j^{(l)}$  fires in a geometrically consistent direction  
 79 across problems in the dataset and is *important* for mathematical reasoning in  $\mathcal{D}$ .  
 80

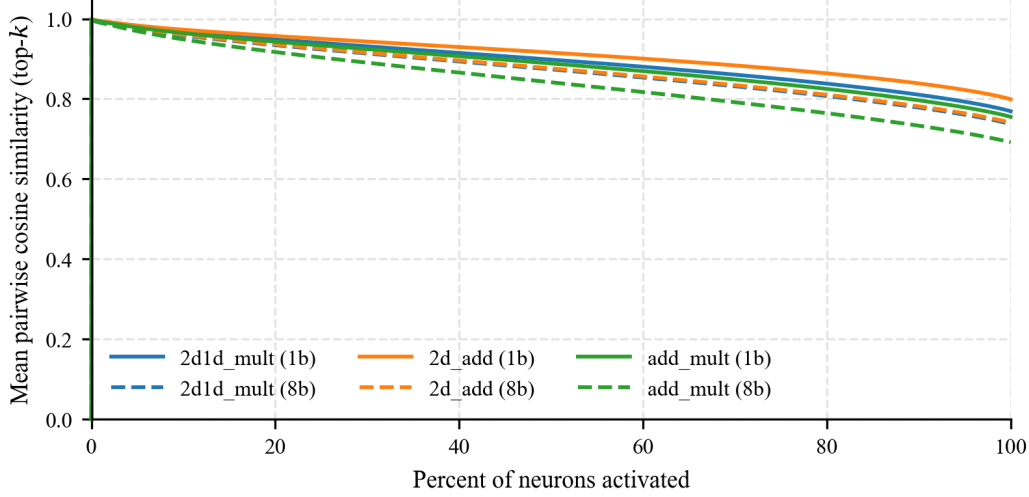


Figure 1: Fraction of Activated Neurons v. Mean Cossine Similarity across Dataset (add, mult, addmult) for 1b and 8b

81 We define the circuit as the set of neurons with the highest cosine similarity scores. Given a  
 82 target activation fraction  $\alpha \in (0, 1]$  we select the top- $k$  neurons globally across all layers where  
 83  $k = \lfloor \alpha \cdot L \cdot d_{\text{ffn}} \rfloor$  resulting in a discovered circuit at sparsity level  $\alpha$ . In Figure 1, we visualize  
 84 the average cosine similarity of the top- $k$  circuit for  $\alpha$  sparsity across various problem classes to  
 85 showcase the tradeoff between fully representing the model and staying relevant to the problem set.

### 86 3.2 K-means Circuit Clustering

87 While the top- $k$  selection identifies which neurons are functionally critical globally, the model may  
 88 route different types of arithmetic problems through distinct computational pathways. To disentangle  
 89 these latent subcircuits, we cluster the arithmetic problems into  $K$  functional classes based on their  
 90 activation trajectories. Let  $a_{i,j}^{(l)}$  denote the activation of neuron  $j$  in layer  $l$  for problem  $i \in \mathcal{D}$ . We  
 91 apply standard  $K$ -means clustering to partition the dataset by minimizing the intra-cluster variance  
 92 of these activations:

$$\min_{\{c_i\}_{i \in \mathcal{D}}, \{\mu_{k,j}^{(l)}\}_{k=1}^K} \sum_{k=1}^K \sum_{\substack{i \in \mathcal{D} \\ c_i=k}} \left\| a_{i,j}^{(l)} - \mu_{k,j}^{(l)} \right\|_2^2 \quad (2)$$

93 where  $c_i$  is the cluster assignment for problem  $i$ , and  $\mu_{k,j}^{(l)}$  is the centroid activation for cluster  $k$ . We  
 94 do not enforce balanced cluster sizes, allowing the natural distribution of problem types to dictate the  
 95 functional density of each latent class.

96 To determine the optimal number of latent circuits ( $K$ ), we analyze the mean cosine distance to the  
 97 cluster centroids as a function of  $K$ . As shown in Figure 2, we utilize the elbow method to select a  $K$   
 98 that balances representational compactness with functional distinctness. Furthermore, we enforce the  
 99 same  $K$  for both the 1B student and 8B teacher models. Establishing a shared  $K$  is mathematically  
 100 essential for our distillation pipeline, as it ensures square similarity matrices and enables the globally  
 101 optimal 1:1 bipartite matching of teacher and student circuits during training (Sec. 3.3). **VG: same K**  
 102 **for 1:1 matching based on what claude said but we can adjust this**

### 103 3.3 Circuit Distillation

104 Our distillation framework trains the student to emulate both the predictive distribution and the  
 105 internal reasoning structures of the teacher. To achieve this, we optimize a composite objective that

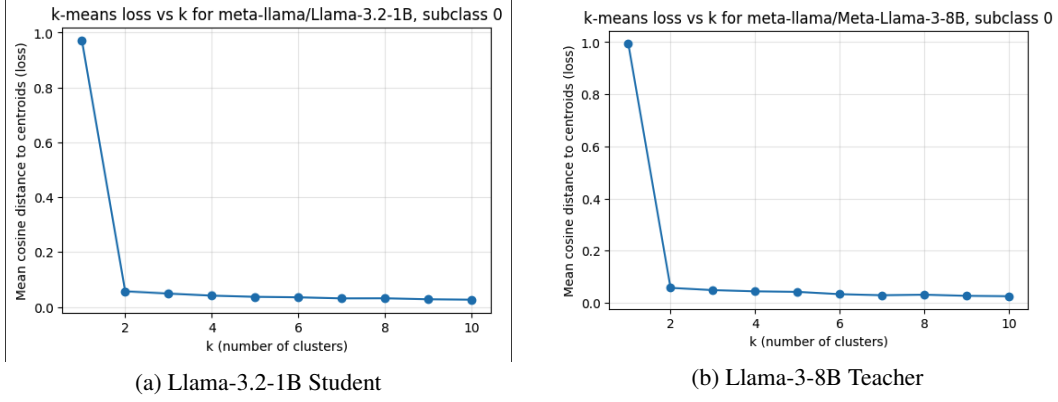


Figure 2: Mean cosine distance to cluster centroids as a function of  $K$  (number of clusters). We apply the elbow method to identify the optimal  $K$ , which is constrained to be equal across both the teacher and student models to facilitate exact 1:1 circuit pairing during distillation.

combines standard Kullback-Leibler (KL) divergence over the output logits with Centered Kernel Alignment (CKA) over the paired latent circuits:

$$\mathcal{L} = \mathcal{L}_{\text{KL}} + \lambda_t \sum_k w_k^{\text{importance}} \cdot w_k^{\text{grad}} \cdot (1 - \text{CKA}_k) \quad (3)$$

where  $w_k^{\text{importance}}$  denotes the task-relevance weight of cluster  $k$ ,  $w_k^{\text{grad}}$  is an intra-circuit normalization term (derived in 3.3.2), and  $\lambda_t$  is an adaptive global weight.

**Adaptive Global Gradient Matching ( $\lambda_t$ ).** A fundamental challenge in multi-objective distillation is the incommensurability of KL divergence (an unbounded statistical measure) and CKA (a bounded geometric similarity in  $[0, 1]$ ). Balancing these terms based on raw loss magnitudes is theoretically flawed, as loss values do not dictate the magnitude of the resulting parameter updates. The mathematically principled approach is gradient norm matching, which defines the optimal trade-off parameter as the ratio of their respective gradients:  $\lambda_t = \|\nabla_{\theta} \mathcal{L}_{\text{KL}}\|_t / \|\nabla_{\theta} \mathcal{L}_{\text{CKA}}\|_t$ .

Crucially, a static  $\lambda$  fixed at initialization is insufficient. As structural alignment improves during training, the CKA gradient naturally shrinks. Because the KL and CKA gradients decay at uncoupled rates, a static weight inevitably becomes miscalibrated, often severely underweighting the mechanistic constraint later in training. However, computing the exact adaptive  $\lambda_t$  at every step requires two independent backward passes, effectively doubling the computational cost. To maintain a strictly calibrated mechanistic constraint without prohibitive overhead, we exploit the analytical structure of the CKA gradient, which scales inversely with the student’s representational energy. We estimate this denominator using  $\mu_1^{(k)}$  computed cheaply during the forward pass, allowing us to dynamically reweight  $\lambda_t$  at minimal cost. Alternatively, exact gradient norms can be periodically recomputed every  $T$  steps to amortize the overhead.

### 3.3.1 Validating Circuit Importance via Invariant Ablation Baselines

Previous macro-level circuit distillation frameworks [Singhal et al., 2025] rely on standard zero-ablation to verify mechanism importance. However, standard ablation fails to account for the fact that removing *any* large set of parameters will natively decrease model accuracy simply by degrading general network capacity. To rigorously validate our identified top- $k$  circuits, we determine an "expected" baseline estimate that isolates specific functional importance from general capacity loss.

We calculate the expected performance drop of zeroing out  $n$  *random* neurons, establishing a baseline that is invariant to the raw number of neurons removed. As shown in Figure 3, ablating our identified top- $k$  circuit (blue) causes a catastrophic drop in mean accuracy at extremely high sparsity levels—zeroing out just 5% of the 1B model reduces accuracy to near zero. Conversely, the expected baseline of randomly ablating the equivalent fraction of neurons (orange) shows negligible impact, requiring the removal of roughly 30% of the network to see comparable degradation. This

138 stark contrast proves that our discovered top- $k$  subset is highly specialized and uniquely responsible  
 139 for arithmetic reasoning, validating it as a robust target for distillation.

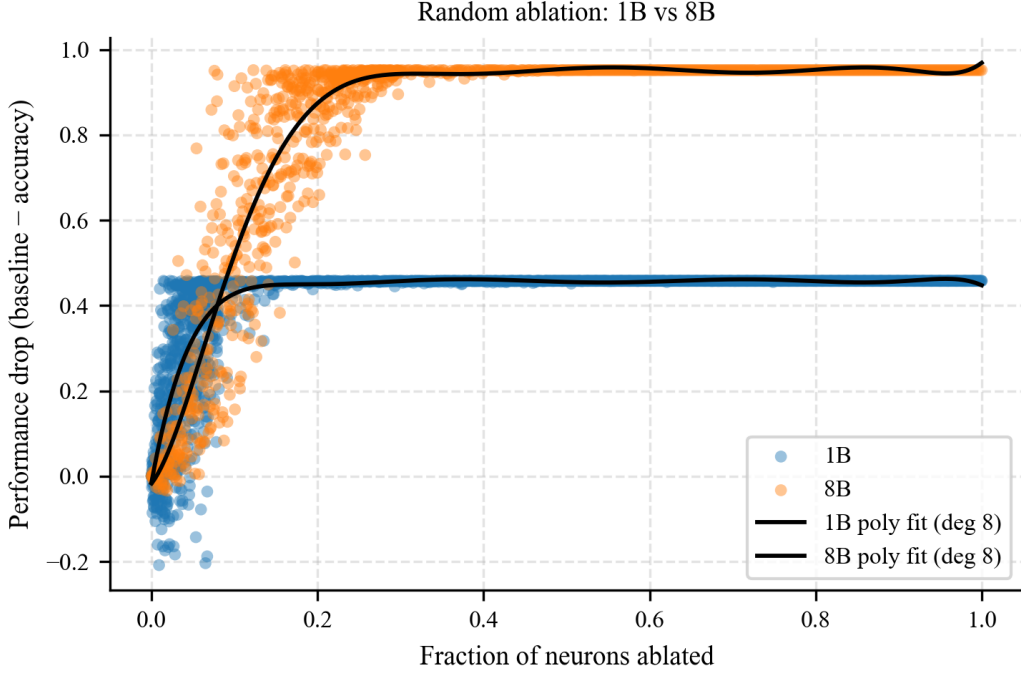


Figure 3: Mean accuracy as a function of the fraction of neurons ablated for Llama-3-1B (left) and Llama-3-8B (right). Ablating the top- $k$  neurons (blue) destroys arithmetic performance almost immediately, whereas ablating random neurons (orange) requires removing over 30% of the network to see comparable degradation.

### 140 3.3.2 Gradient-Based Intra-Circuit Weighting ( $w_k^{\text{grad}}$ )

141 A naive unweighted sum over clusters in the CKA loss assumes that all functional units exert equal  
 142 gradient pressure. However, our analysis reveals that highly sparse, functionally specialized clusters  
 143 often exhibit a stable rank near 1 ( $\rho^{(k)} \approx 1$ ), meaning the neurons within the cluster collectively  
 144 encode a single primary direction.

145 The CKA loss for cluster  $k$  is  $\mathcal{L}_k = 1 - \text{CKA}(X_k, Y_k)$ . Having empirically shown  $\rho^{(k)} \approx 1$ , both  
 146 the student and teacher Gram matrices are approximately rank-1. Therefore,  $\text{tr}(K_c^{(k)2}) \approx \mu_1^{(k)2}$   
 147 and  $\text{tr}(L_c^{(k)2}) \approx \lambda_1^{(k)2}$ , where  $\mu_1^{(k)}$  and  $\lambda_1^{(k)}$  are the leading eigenvalues of the student and teacher  
 148 centered Gram matrices, respectively.

149 The gradient with respect to the student activation matrix  $X_k$  reduces to:

$$\frac{\partial \mathcal{L}_k}{\partial X_k} \approx \frac{-2 H L_c^{(k)} H X_k}{\sqrt{\text{tr}(K_c^{(k)2}) \cdot \text{tr}(L_c^{(k)2})}}. \quad (4)$$

150 Consequently, the gradient norm scales inversely with the student’s leading eigenvalue:

$$\left\| \frac{\partial \mathcal{L}_k}{\partial X_k} \right\| \sim \frac{\lambda_1^{(k)}}{\mu_1^{(k)} \cdot \lambda_1^{(k)}} = \frac{1}{\mu_1^{(k)}}. \quad (5)$$

151 Because the teacher eigenvalue  $\lambda_1^{(k)}$  cancels out, the gradient is suppressed purely by the scale of the  
 152 *student* activations. Larger clusters yield disproportionately smaller per-neuron parameter updates

Table 1: FLOPs analysis comparing standard distillation to circuit distillation on 2 digit addition

|                | Standard | 0.01, constant $\lambda$ | 0.01, adaptive $\lambda$ | 0.10, adaptive $\lambda$ |
|----------------|----------|--------------------------|--------------------------|--------------------------|
| PFLOPS / Epoch | 1.077    | 1.078                    | 1.259                    | 1.440                    |

153 simply because their energy  $\mu_1^{(k)}$  is larger. This is an artifact of scale, not a meaningful learning  
 154 signal.

155 To equalize gradient contributions across clusters, we multiply each loss term by  $\mu_1^{(k)}$  so that  
 156  $w_k^{\text{grad}} \cdot \|\partial \mathcal{L}_k / \partial X_k\|$  remains constant across  $k$ . Normalizing so the total correction is invariant to  
 157 circuit size yields the final per-cluster gradient correction weight:

$$w_k^{\text{grad}} = \frac{\mu_1^{(k)}}{\sum_j \mu_1^{(j)}} \quad (6)$$

158 This ensures that an equivalent degree of structural misalignment produces a commensurate optimiza-  
 159 tion update, independent of the cluster’s raw neuron count.

## 160 4 Experiments

### 161 4.1 $\lambda$ Testing

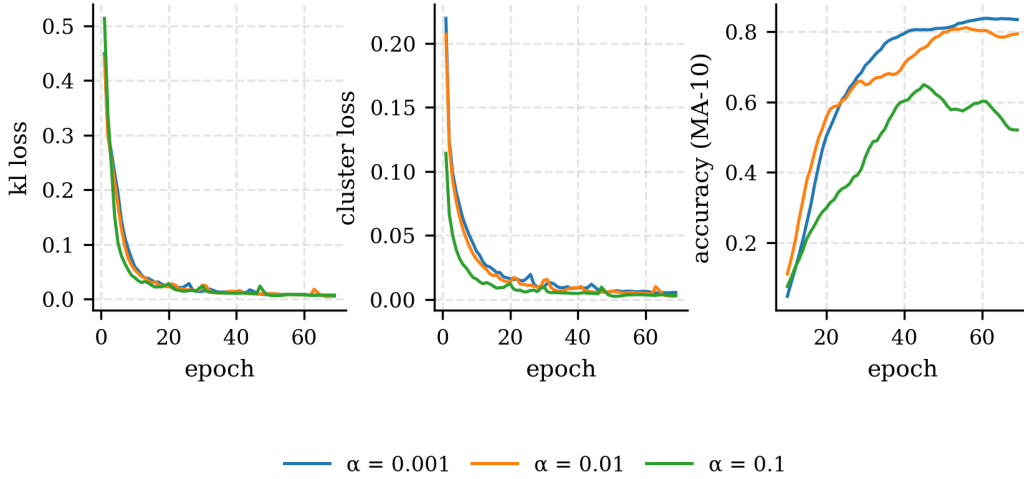


Figure 4:  $\alpha_{testing}$

### 162 4.2 $\alpha$ Testing

### 163 4.3 FLOPs Analysis

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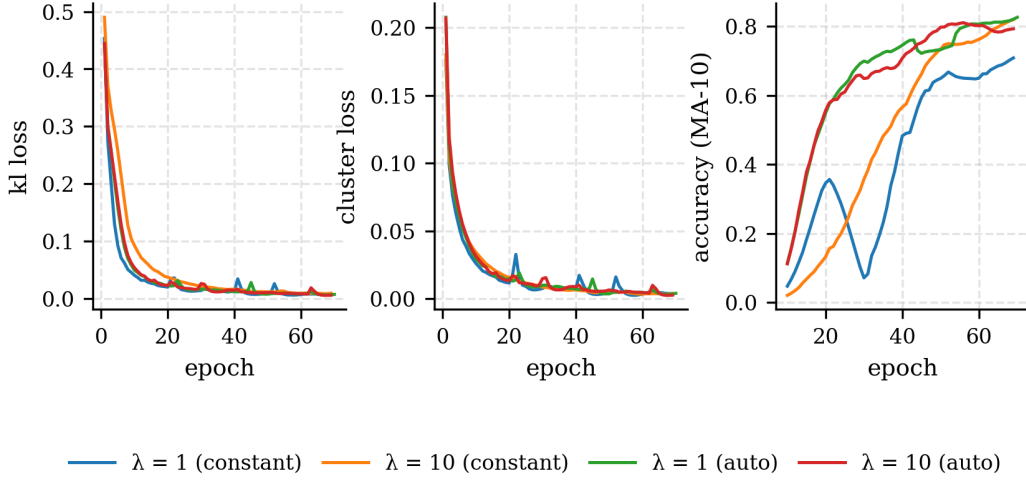


Figure 5:  $t_{\text{esting}}$

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